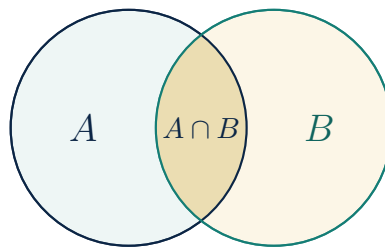


# Sets, Relations & Functions



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Roll: 43 • Batch: CSE-19 • Session: 2025–2026

**COURSE:**

**CSE 1143: Discrete Mathematics (3.00 Credits)**

**DEPARTMENT:**

Dept. of Computer Science & Engineering

**INSTITUTION:**

**Sylhet Engineering College (SEC)**

Affiliated with Shahjalal University of Science & Technology (SUST)

## ★ SEC TT-01 Tutorial Test Syllabus Guide

**Tutorial Test Date: 7 September, 2026 • Dept. of Computer Science & Engineering**

Topics marked with a star (★) are explicitly included in your upcoming TT-01 syllabus: *Empty / Null Set, Equal Set, Cardinality, Power Set, Set Operations, Venn Diagrams, Domain & Range, and Reflexive Relations.*

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# 1 The Intuitive Foundation of Sets

## 1.1 Everyday Mental Model: The Shopping Basket

Imagine walking through a supermarket with a shopping basket.

- You put in an apple, a mango, and a banana. The order in which you tossed them into the basket does not change what you bought.
- If you accidentally scan the exact same bottle of milk twice on a checklist, you still only own distinct items of that kind in your mathematical universe.
- Everything in your basket is strictly defined: either an item is in the basket, or it is not.

### Formal Definition: Set

A **set** is a well-defined collection of distinct objects. The objects belonging to a set are called its **elements** or **members**.

- If  $x$  is an element of set  $A$ , we write:  $x \in A$  (read: “ $x$  belongs to  $A$ ”).
- If  $x$  is not an element of set  $A$ , we write:  $x \notin A$  (read: “ $x$  does not belong to  $A$ ”).

### The Two Fundamental Axioms of Elementary Sets

1. **Order Invariance:**  $\{1, 2, 3\} = \{3, 1, 2\} = \{2, 1, 3\}$ . Sets do not have a first, second, or last element.
2. **Duplication Invariance:** Repeated elements are redundant:  $\{1, 1, 2, 3, 3, 3\} = \{1, 2, 3\}$ .

## 1.2 Set Representations

In computer science and mathematics, we specify sets in two primary ways:

### 1.2.1 1. Roster / Listing Form

All elements are explicitly written out inside curly braces separated by commas:

$$A = \{2, 4, 6, 8, 10\} \quad \text{or} \quad V = \{a, e, i, o, u\}$$

For infinite or large patterns, an ellipsis (...) indicates continuation:  $E = \{2, 4, 6, 8, \dots\}$ .

### 1.2.2 2. Set-Builder / Predicate Notation

Instead of listing elements, we state the exact mathematical rule (predicate  $P(x)$ ) that any candidate element  $x$  must satisfy to qualify:

$$A = \{x \in U \mid P(x)\} \quad \text{or} \quad A = \{x : P(x)\}$$

*Read as:* “The set of all  $x$  such that property  $P(x)$  is true.”

- Example 1:  $B = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\} \implies B = \{1, 3, 5, 7, 9\}$ .
- Example 2:  $S = \{x \in \mathbb{R} \mid x^2 - 4 = 0\} \implies S = \{-2, 2\}$ .

## 1.3 Standard Universal Number Systems

Throughout discrete mathematics and competitive programming, the following blackboard bold symbols are standardized:

| Symbol         | Set Name          | Standard Definition / Elements                                   |
|----------------|-------------------|------------------------------------------------------------------|
| $\mathbb{N}$   | Natural Numbers   | $\{0, 1, 2, 3, \dots\}$ (Note: In CS/Rosen, $0 \in \mathbb{N}$ ) |
| $\mathbb{Z}$   | Integers          | $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$                       |
| $\mathbb{Z}^+$ | Positive Integers | $\{1, 2, 3, 4, \dots\}$                                          |
| $\mathbb{Q}$   | Rational Numbers  | $\{p/q \mid p, q \in \mathbb{Z} \text{ and } q \neq 0\}$         |
| $\mathbb{R}$   | Real Numbers      | All rational and irrational numbers (the continuous real line)   |
| $\mathbb{C}$   | Complex Numbers   | $\{a + bi \mid a, b \in \mathbb{R} \text{ and } i = \sqrt{-1}\}$ |

### [SEC Final Examination 2022, CSE 103] Set Equality, Element Listing & Finiteness

#### QUESTION

- Which of these sets are equal:  $S_1 = \{x, y, z\}$ ,  $S_2 = \{z, y, z, x\}$ ,  $S_3 = \{y, x, y, z\}$ ,  $S_4 = \{y, z, x, y\}$ ? [2]
- List the elements of each set where  $\mathbb{N} = \{1, 2, 3, \dots\}$ : (i)  $A = \{x \in \mathbb{N} \mid 3 < x < 9\}$  (ii)  $B = \{x \in \mathbb{N} \mid x \text{ is even, } x < 11\}$  (iii)  $C = \{x \in \mathbb{N} \mid 4 + x = 3\}$ . [1×3]
- Determine which of the following sets are finite: (i) Lines parallel to the  $x$ -axis (ii) Letters in the English alphabet (iii) Integers which are multiples of 5 (iv) Animals living on the earth. [2]

#### ANSWER

- By Order/Duplication Invariance, all repeat and reorder to  $\{x, y, z\} \implies \mathbf{S_1 = S_2 = S_3 = S_4}$ .
- $\mathbf{A = \{4, 5, 6, 7, 8\}; B = \{2, 4, 6, 8, 10\}; C : x = -1 \notin \mathbb{N} \implies C = \emptyset}$ .
- (i) **Infinite** (ii) **Finite** ( $|S| = 26$ ) (iii) **Infinite** (iv) **Finite** (population is bounded at any instant).

## 2 Special Sets, Subsets & The Great Trap: $\in$ vs. $\subseteq$

### 2.1 ★ Special Types of Sets: Empty / Null Set & Universal Set

★ **TT-01 Syllabus Topic (7 Sept 2026):** Empty Set or Null Set ( $\emptyset$ )

- Universal Set ( $\mathcal{U}$  or  $U$ ):** The overarching domain containing all possible objects under current discussion.
- Empty Set / Null Set ( $\emptyset$  or  $\{\}$ ):** A set containing zero elements ( $|\emptyset| = 0$ ).
- Singleton Set:** Any set containing exactly one single element, e.g.,  $\{42\}$  or  $\{\emptyset\}$ .

### [SEC TT-01 / Final Examination 2024, CSE 0541 1143] Set Definition & Classification of Sets

#### QUESTION

Define a set and list the different types of sets. Give an example for each type.

[2+4]

#### ANSWER

A set is a well-defined collection of distinct objects (elements).

- Finite Set:** Contains a countable number of elements. Example:  $A = \{2, 4, 6\}$ ,  $|A| = 3$ .
- Infinite Set:** Elements cannot be exhausted. Example:  $\mathbb{N} = \{1, 2, 3, \dots\}$ .
- Empty / Null Set ( $\emptyset$ ):** Contains zero elements ( $|\emptyset| = 0$ ). Example:  $\{x \in \mathbb{R} \mid x^2 + 1 = 0\} = \emptyset$ .
- Singleton Set:** Contains exactly one element. Example:  $\{0\}$  or  $\{\emptyset\}$ .
- Universal Set ( $U$ ):** Contains all objects under discussion. Example:  $U = \mathbb{R}$ .
- Equal Sets ( $A = B$ ):** Contain identical members. Example:  $\{1, 2\} = \{2, 1, 1\}$ .

- **Subset** ( $A \subseteq B$ ):  $\forall x(x \in A \implies x \in B)$ . Example:  $\{1, 2\} \subseteq \{1, 2, 3\}$ .
- **Power Set** ( $\mathcal{P}(S)$ ): Set of all subsets. Example:  $\mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}$ .

### The Classic Empty Set Trap: $\emptyset$ vs. $\{\emptyset\}$

A common beginner mistake is confusing the empty set  $\emptyset$  with the singleton set  $\{\emptyset\}$ .

- $\emptyset$ : An empty plastic box. It contains zero items inside. Cardinality  $|\emptyset| = 0$ .
- $\{\emptyset\}$ : A plastic box that has an empty plastic box placed inside it. It contains **one object** (which happens to be an empty box!). Cardinality  $|\{\emptyset\}| = 1$ .
- Therefore:  $\emptyset \neq \{\emptyset\}$  and  $|\emptyset| \neq |\{\emptyset\}|$ .

## 2.2 ★ Subsets, Strict Subsets and Equal Sets

★ **TT-01 Syllabus Topic (7 Sept 2026):** *Equal Sets* ( $A = B \iff A \subseteq B \wedge B \subseteq A$ )

### Subset ( $\subseteq$ ) and Proper Subset ( $\subset$ )

Let  $A$  and  $B$  be two arbitrary sets:

- $A \subseteq B$  ( $A$  is a subset of  $B$ ) if and only if every element of  $A$  is also an element of  $B$ :

$$A \subseteq B \iff \forall x(x \in A \implies x \in B)$$

- $A \subset B$  or  $A \subsetneq B$  ( $A$  is a **proper** or **strict** subset of  $B$ ) if  $A \subseteq B$  and  $A \neq B$  (i.e.,  $B$  contains at least one element not present in  $A$ ).
- **Set Equality** ( $A = B$ ): Two sets are equal if and only if they contain the exact same elements:

$$A = B \iff (A \subseteq B \wedge B \subseteq A)$$

### Theorem: The Empty Set is a Subset of Every Set

**Proposition:** For any arbitrary set  $S$ ,  $\emptyset \subseteq S$ .

**Proof (Vacuous Truth):** By definition of subset,  $\emptyset \subseteq S$  means:

$$\forall x(x \in \emptyset \implies x \in S)$$

Because the empty set  $\emptyset$  has no elements, the premise  $x \in \emptyset$  is **always False** for every possible  $x$ . In classical logic, any conditional statement with a False antecedent ( $F \implies P$ ) is **vacuously True**. Thus,  $\emptyset \subseteq S$  holds universally for every set  $S$ . ■

## 2.3 The Membership ( $\in$ ) vs. Inclusion ( $\subseteq$ ) Dissection

The distinction between element membership and subset inclusion represents one of the most common conceptual pitfalls where students make errors.

### The Gold Rule of $\in$ vs. $\subseteq$

- $x \in A$  is a relationship between an **individual element** and a set.  $x$  must literally appear as an item inside  $A$ 's braces.
- $S \subseteq A$  is a relationship between **two sets**. Every element inside  $S$  must belong to  $A$ .
- **The Braces Conversion Rule:** If  $x \in A$ , then wrapping  $x$  in a set gives  $\{x\} \subseteq A$ .

**Master Example: Analyzing  $A = \{1, 2, \{3\}, \{4, 5\}, \emptyset\}$** 

Consider the set:  $A = \{1, 2, \{3\}, \{4, 5\}, \emptyset\}$ .

| Statement                   | Truth Value  | Rigorous Mathematical Explanation                                           |
|-----------------------------|--------------|-----------------------------------------------------------------------------|
| $1 \in A$                   | <b>True</b>  | 1 is an explicitly listed member of $A$ .                                   |
| $\{1\} \in A$               | <b>False</b> | $\{1\}$ is not in the set; only the naked integer 1 is present.             |
| $\{1\} \subseteq A$         | <b>True</b>  | The single element inside $\{1\}$ , namely 1, belongs to $A$ .              |
| $3 \in A$                   | <b>False</b> | 3 is locked inside a nested subset; it is not a direct member.              |
| $\{3\} \in A$               | <b>True</b>  | The set $\{3\}$ is itself a direct element inside $A$ .                     |
| $\{3\} \subseteq A$         | <b>False</b> | The element inside $\{3\}$ is 3, but $3 \notin A$ .                         |
| $\{\{3\}\} \subseteq A$     | <b>True</b>  | The element inside $\{\{3\}\}$ is $\{3\}$ , and $\{3\} \in A$ .             |
| $\emptyset \in A$           | <b>True</b>  | $\emptyset$ is explicitly written inside $A$ as an element.                 |
| $\emptyset \subseteq A$     | <b>True</b>  | The empty set is vacuously a subset of every set.                           |
| $\{\emptyset\} \subseteq A$ | <b>True</b>  | The element inside $\{\emptyset\}$ is $\emptyset$ , and $\emptyset \in A$ . |

### 3 Cardinality & The Power Set Explosion

#### 3.1 ★ Cardinality of a Set

★ **TT-01 Syllabus Topic (7 Sept 2026):** *Cardinality of a Set ( $|S| = n$ )*

##### Cardinality

If  $S$  is a finite set containing  $n$  distinct elements (where  $n \in \mathbb{N}$ ), we say  $S$  is a finite set with **cardinality**  $n$ , denoted by  $|S| = n$ .

- $|\emptyset| = 0$
- $|\{a, b, c\}| = 3$
- $|\{1, \{2, 3\}, 4\}| = 3$  (the pair  $\{2, 3\}$  counts as 1 single composite element).

##### [SEC Final Examination 2022, CSE 103] Cardinal Number of Sets

##### QUESTION

Find the cardinal number of each of the following sets: (i)  $A = \{a, b, c, \dots, y, z\}$  (ii)  $B = \{x \mid x \in \mathbb{N}, x^2 = 5\}$  (iii)  $C = \{10, 20, 30, 40, \dots\}$  (iv)  $D = \{6, 7, 8, 9, \dots\}$ . [2]

##### ANSWER

- $A = 26$  lower-case letters  $\implies |A| = 26$ .
- $x^2 = 5 \implies x = \pm\sqrt{5} \notin \mathbb{N} \implies B = \emptyset \implies |B| = 0$ .
- $C = \{10k \mid k \in \mathbb{Z}^+\} \implies |C| = \aleph_0$  (Countably Infinite).
- $D = \{n \in \mathbb{N} \mid n \geq 6\} \implies |D| = \aleph_0$  (Countably Infinite).

#### 3.2 ★ The Power Set ( $\mathcal{P}(S)$ ) & Cardinality $2^n$

★ **TT-01 Syllabus Topic (7 Sept 2026):** *Power Set and Cardinality Theorem  $|\mathcal{P}(S)| = 2^n$*

### Power Set

Given a set  $S$ , the **power set** of  $S$ , denoted by  $\mathcal{P}(S)$  or  $2^S$ , is the set of **all possible subsets** of  $S$ :

$$\mathcal{P}(S) = \{X \mid X \subseteq S\}$$

### Theorem: Cardinality of the Power Set

**Proposition:** If  $|S| = n$ , then  $|\mathcal{P}(S)| = 2^n$ .

**Intuitive Proof (The Binary Decision Tree):**

To construct any arbitrary subset  $X \subseteq S$ , we inspect each of the  $n$  elements in  $S$  one by one. For each element  $x_i$ , we have exactly **2 choices**:

$$\text{Choice: } \begin{cases} \text{Include } x_i \text{ in } X & (\text{Bit 1}) \\ \text{Exclude } x_i \text{ from } X & (\text{Bit 0}) \end{cases}$$

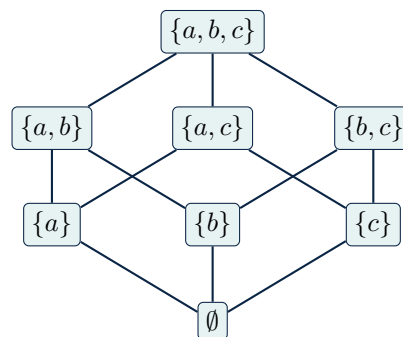
By the Fundamental Counting Rule of Combinatorics:

$$\underbrace{2 \times 2 \times 2 \times \cdots \times 2}_{n \text{ decisions}} = 2^n \text{ unique subsets.} \quad \blacksquare$$

### Tracing All Subsets of $S = \{a, b, c\}$

Let  $S = \{a, b, c\}$ . Since  $|S| = 3$ ,  $|\mathcal{P}(S)| = 2^3 = 8$  subsets:

$$\mathcal{P}(S) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$



**Figure 1:** Hasse subset-inclusion lattice for the Boolean algebra  $\mathcal{P}(\{a, b, c\})$ .

### 3.3 Recursive Power Sets of the Null Set

What happens when we repeatedly apply the power set operator to the empty set?

1. Start with  $\emptyset$ :  $|\emptyset| = 0$ .
2.  $\mathcal{P}(\emptyset) = \{\emptyset\}$ . Cardinality  $= 2^0 = 1$ .
3.  $\mathcal{P}(\mathcal{P}(\emptyset)) = \mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$ . Cardinality  $= 2^1 = 2$ .
4.  $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset))) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$ . Cardinality  $= 2^2 = 4$ .
5.  $|\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset))))| = 2^4 = 16$ .

## 4 Cartesian Product & Ordered $n$ -Tuples

## 4.1 Ordered Pairs vs. Unordered Sets

In a set, order is ignored:  $\{1, 2\} = \{2, 1\}$ . But in computer programming (such as coordinates, database tables, or key-value tuples), position matters! An **ordered pair**  $(a, b)$  has a designated first element  $a$  and second element  $b$ :

$$(a, b) = (c, d) \iff (a = c \wedge b = d)$$

### Cartesian Product $(A \times B)$

The **Cartesian product** of two sets  $A$  and  $B$ , denoted by  $\mathbf{A} \times \mathbf{B}$ , is the set of all ordered pairs  $(a, b)$  where  $a \in A$  and  $b \in B$ :

$$\mathbf{A} \times \mathbf{B} = \{(a, b) \mid a \in \mathbf{A} \wedge b \in \mathbf{B}\}$$

### Properties of the Cartesian Product

- **Cardinality Rule:** If  $|A| = m$  and  $|B| = n$ , then:  $|\mathbf{A} \times \mathbf{B}| = \mathbf{m} \cdot \mathbf{n}$ .
- **Non-Commutative:** In general,  $\mathbf{A} \times \mathbf{B} \neq \mathbf{B} \times \mathbf{A}$  (unless  $A = B$  or one of them is  $\emptyset$ ).
- **Null Absorption:**  $\mathbf{A} \times \emptyset = \emptyset \times \mathbf{A} = \emptyset$  (since no ordered pair can be formed with an element from  $\emptyset$ ).

### Cartesian Product Calculation

Let  $A = \{1, 2\}$  and  $B = \{x, y, z\}$ .

- $A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$ . Cardinality:  $2 \times 3 = 6$ .
- $B \times A = \{(x, 1), (x, 2), (y, 1), (y, 2), (z, 1), (z, 2)\}$ .
- Observe that  $(1, x) \neq (x, 1)$ , confirming  $A \times B \neq B \times A$ .

### [SEC TT-01, CSE 103, 2022] Cartesian Product, Set Intersection & Floor/Ceiling

#### QUESTION

- (a) Let  $A = \{1, 2, 3\}$ ,  $B = \{a, b, c, d\}$  and  $C = \{c, d\}$ . Find:  $(A \times B) \cap (A \times C)$  and  $(B \cap C)$ . [5]  
 (b) Find: (i)  $\lfloor 7.5 \rfloor$ ,  $\lfloor -7.5 \rfloor$ ,  $\lfloor -18 \rfloor$ ; (ii)  $\lceil 7.5 \rceil$ ,  $\lceil -7.5 \rceil$ ,  $\lceil -18 \rceil$ . [5]

#### ANSWER

- (a)  $B \cap C = \{c, d\}$ . By distributivity,  $(A \times B) \cap (A \times C) = A \times (B \cap C)$

$$= \{(1, c), (1, d), (2, c), (2, d), (3, c), (3, d)\}$$

- (b)  $\lfloor 7.5 \rfloor = 7$ ,  $\lfloor -7.5 \rfloor = -8$ ,  $\lfloor -18 \rfloor = -18$ ;  $\lceil 7.5 \rceil = 8$ ,  $\lceil -7.5 \rceil = -7$ ,  $\lceil -18 \rceil = -18$ .

## 4.2 ★ Relations Introduction: Domain, Range & Reflexive Relations

★ TT-01 Syllabus Topic (7 Sept 2026): *Binary Relations, Domain, Range, and Reflexivity*

### Binary Relation, Domain & Range

Let  $A$  and  $B$  be sets. A **binary relation**  $R$  from  $A$  to  $B$  is any subset of the Cartesian product  $A \times B$ :

$$R \subseteq A \times B$$

If  $(a, b) \in R$ , we write  $a R b$  (read: “ $a$  is related to  $b$ ”). If  $A = B$ ,  $R$  is called a relation on  $A$ .

- **Domain of  $R$  ( $\text{Dom}(R)$ ):** The set of all first coordinates from the ordered pairs in  $R$ :

$$\text{Dom}(R) = \{a \in A \mid \exists b \in B, (a, b) \in R\} \subseteq A$$

- **Range of  $R$  ( $\text{Range}(R)$ ):** The set of all second coordinates from the ordered pairs in  $R$ :

$$\text{Range}(R) = \{b \in B \mid \exists a \in A, (a, b) \in R\} \subseteq B$$

### Core Relation Properties: Reflexive, Symmetric, Transitive & Equivalence

Let  $R$  be a binary relation on a set  $A$ :

- **Reflexive:**  $\forall a \in A, (a, a) \in R$ . (Matrix diagonal  $m_{ii} = 1$ ; Digraph has self-loops on all vertices).
- **Symmetric:**  $\forall x, y \in A, (x, y) \in R \implies (y, x) \in R$ . (Matrix satisfies  $M = M^T$ ).
- **Antisymmetric:**  $\forall x, y \in A, ((x, y) \in R \wedge (y, x) \in R) \implies x = y$ .
- **Transitive:**  $\forall x, y, z \in A, ((x, y) \in R \wedge (y, z) \in R) \implies (x, z) \in R$ .
- **Equivalence Relation:** A relation  $R$  that is simultaneously **reflexive**, **symmetric**, and **transitive**.

### [SEC Final Examination 2023, CSE 143] Relation Matrix, Arrow Diagram, Inverse, Domain & Range

#### QUESTION

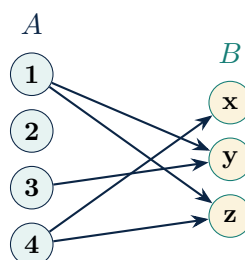
Given  $A = \{1, 2, 3, 4\}$  and  $B = \{x, y, z\}$ . Let  $R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$  be a relation from  $A$  to  $B$ . (i) Determine the matrix of the relation. (ii) Draw the arrow diagram of  $R$ . (iii) Find the inverse relation  $R^{-1}$ . (iv) Determine the domain and range of  $R$ . [4]

#### ANSWER

(i) **Matrix** (rows  $A = (1, 2, 3, 4)$ , columns  $B = (x, y, z)$ ):

$$M_R = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

(ii) **Arrow Diagram:**



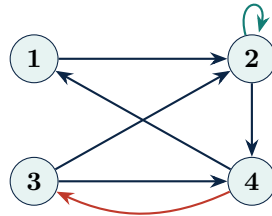
(iii)  $R^{-1} = \{(y, 1), (z, 1), (y, 3), (x, 4), (z, 4)\}$ . (iv)  $\text{Dom}(R) = \{1, 3, 4\}$ ,  $\text{Range}(R) = \{x, y, z\}$ .

### [SEC Final Examination 2023, CSE 143] Digraph Model of a Binary Relation

#### QUESTION

Define Relation. Find the relation  $R$  determined by the directed graph below:

[1+2]



#### ANSWER

**Definition of Relation:** A binary relation  $R$  from set  $A$  to set  $B$  is any mathematical subset of the Cartesian product  $A \times B$  ( $R \subseteq A \times B$ ). When  $A = B$ ,  $R$  is a relation on  $A$ .

**Extracting Directed Edges into Ordered Pairs:** We inspect the directed arrows emanating from each vertex in vertex set  $V = \{1, 2, 3, 4\}$ :

- From vertex 1: edge to 2  $\implies (1, 2)$ .
- From vertex 2: self-loop at 2 and edge to 4  $\implies (2, 2), (2, 4)$ .
- From vertex 3: edges to 2 and 4  $\implies (3, 2), (3, 4)$ .
- From vertex 4: edges to 1 and 3 (curved)  $\implies (4, 1), (4, 3)$ .

Thus, the binary relation represented by the directed graph is:

$$R = \{(1, 2), (2, 2), (2, 4), (3, 2), (3, 4), (4, 1), (4, 3)\} \quad (|R| = 7 \text{ pairs}).$$

### [SEC Final Examination 2023, CSE 143] Reflexive, Symmetric & Transitive Closures

#### QUESTION

Consider the relation  $R = \{(a, a), (a, b), (b, c), (c, c)\}$  on the set  $A = \{a, b, c\}$ . Find: (i)  $\text{reflexive}(R)$ ; (ii)  $\text{symmetric}(R)$ ; (iii)  $\text{transitive}(R)$ . [3]

#### ANSWER

The **closure** of a relation  $R$  with respect to property  $P$  is the smallest relation  $R' \supseteq R$  that satisfies property  $P$ :

- (i) **Reflexive Closure** ( $\text{reflexive}(R) = R \cup \Delta_A$ ): A relation on  $A = \{a, b, c\}$  is reflexive iff it contains the full diagonal  $\Delta_A = \{(a, a), (b, b), (c, c)\}$ . In  $R$ , the self-loop  $(b, b)$  is missing. Adding it yields:

$$\text{reflexive}(R) = \{(a, a), (a, b), (b, b), (b, c), (c, c)\}$$

- (ii) **Symmetric Closure** ( $\text{symmetric}(R) = R \cup R^{-1}$ ): We must add  $(y, x)$  for every  $(x, y) \in R$ . The pairs  $(a, b)$  and  $(b, c)$  lack reverse pairs  $(b, a)$  and  $(c, b)$ :

$$\text{symmetric}(R) = \{(a, a), (a, b), (b, a), (b, c), (c, b), (c, c)\}$$

- (iii) **Transitive Closure** ( $\text{transitive}(R) = R^+$ ): We identify all path chains  $(x, y) \in R \wedge (y, z) \in R \implies (x, z) \in R$ . Here,  $(a, b) \in R$  and  $(b, c) \in R$ , which requires the transitive shortcut pair  $(a, c)$ . No further composite chains exist:

$$\text{transitive}(R) = \{(a, a), (a, b), (a, c), (b, c), (c, c)\}$$

[SEC TT-01 / Final Examination 2024, CSE 0541 1143] Tracing Relations on  $A = \{1, 2, 3, 4\}$

### QUESTION

Consider the five relations on  $A = \{1, 2, 3, 4\}$ :  $R = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$ ;  $S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$ ;  $T = \{(1, 3), (2, 1)\}$ ;  $U = A \times A$ ;  $V = \emptyset$ . Determine with reason whether each is (i) Reflexive (ii) Symmetric (iii) Transitive. Define Equivalence relation with example. [6+2]

### ANSWER

- **R**: Reflexive — **No** (missing  $(2, 2), (3, 3)$ ). Symmetric — **No** ( $(1, 2) \in R, (2, 1) \notin R$ ). Transitive — **Yes**.
- **S**: Reflexive — **Yes**. Symmetric — **Yes**. Transitive — **Yes**  $\implies$  **Equivalence Relation**.
- **T**: Reflexive — **No** (no self-loops). Symmetric — **No**. Transitive — **Yes** (vacuously; no chaining pairs).
- **U** =  $A \times A$ : Reflexive, Symmetric, Transitive — **all Yes** (contains every pair).
- **V** =  $\emptyset$ : Reflexive — **No** (since  $A \neq \emptyset$ ). Symmetric, Transitive — **Yes** (vacuously true).

**Equivalence Relation:**  $R$  is an equivalence relation iff reflexive, symmetric, *and* transitive — e.g., congruence mod  $m$  on  $\mathbb{Z}$ , or  $S$  above.

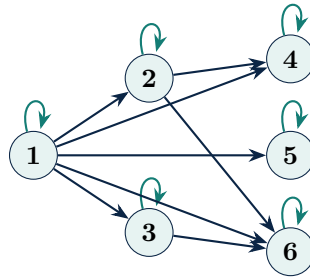
## [SEC TT-01 / Final Examination 2024, CSE 0541 1143] Divisibility Relation &amp; Directed Graph

## QUESTION

Let  $A = \{1, 2, 3, 4, 5, 6\}$  and let  $R$  be the relation on  $A$ :  $R = \{(x, y) \mid x \text{ divides } y\}$ . (i) Write  $R$  as a set of ordered pairs. (ii) Draw its directed graph. [2+3]

## ANSWER

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\} \quad (|R| = 14)$$



## [SEC TT-02, CSE 103, 2022] Matrix Representation: Reflexive, Symmetric &amp; Antisymmetric

## QUESTION

Determine whether the relation represented by the matrix below is reflexive, symmetric, or anti-symmetric?

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

[3]

## ANSWER

**Reflexive:** all diagonal entries  $m_{11} = m_{22} = m_{33} = m_{44} = 1 \implies$  **Yes**. **Symmetric:**  $m_{12} = 1$  but  $m_{21} = 0 \implies M \neq M^T \implies$  **No**. **Antisymmetric:** every off-diagonal 1 ( $m_{12}, m_{13}, m_{34}, m_{41}$ ) has its mirror entry 0  $\implies$  **Yes**.

## [SEC TT-02, CSE 103, 2022] Function vs. Relation

## QUESTION

Is every function a relation and vice versa? [3]

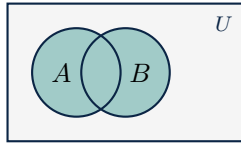
## ANSWER

**Every function is a relation** —  $f : A \rightarrow B$  is simply  $R \subseteq A \times B$  where each  $a \in A$  pairs with exactly *one*  $b \in B$ . **Not every relation is a function** — a relation may map one  $a$  to several  $b$ 's (e.g.,  $\{(1, 2), (1, 3)\}$ ) or leave some  $a$  unassigned.

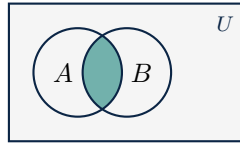
## 5 ★ Fundamental Set Operations & Venn Diagrams

★ TT-01 Syllabus Topic (7 Sept 2026): *Set Operations and Venn Diagrams*

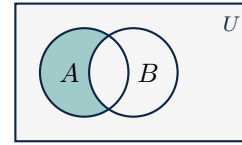
Venn diagrams provide a visual geometry for Boolean set logic inside a bounding Universal set  $\mathcal{U}$ .



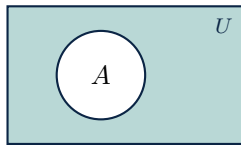
(a) **Union:**  $A \cup B$



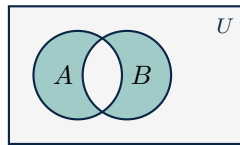
(b) **Intersection:**  $A \cap B$



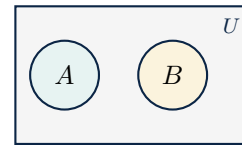
(c) **Difference:**  $A \setminus B$



(d) **Complement:**  $\bar{A}$



(e) **Symm. Difference:**  $A \oplus B$



(f) **Disjoint Sets:**  $A \cap B = \emptyset$

### 5.1 The 5 Core Operations Formalized

#### Mathematical Definitions of Set Operations

1. **Union** ( $A \cup B$ ): Elements belonging to **A or B** (or both):

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

2. **Intersection** ( $A \cap B$ ): Elements belonging to **both** A and B:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

If  $A \cap B = \emptyset$ , sets A and B are said to be **mutually disjoint**.

3. **Set Difference / Relative Complement** ( $A \setminus B$  or  $A - B$ ): Elements in A that are **not** in B:

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\} = A \cap \bar{B}$$

4. **Absolute Complement** ( $\bar{A}$  or  $A^c$ ): Elements in the Universal set  $\mathcal{U}$  that do not belong to A:

$$\bar{A} = \mathcal{U} \setminus A = \{x \in \mathcal{U} \mid x \notin A\}$$

5. **Symmetric Difference** ( $A \oplus B$  or  $A \Delta B$ ): Elements belonging to A or B, but **not both** (The XOR of Sets):

$$\mathbf{A} \oplus \mathbf{B} = (\mathbf{A} \setminus \mathbf{B}) \cup (\mathbf{B} \setminus \mathbf{A}) = (\mathbf{A} \cup \mathbf{B}) \setminus (\mathbf{A} \cap \mathbf{B})$$

#### [SEC TT-01, CSE (DM), year not printed on paper] Elementary Set Operations

##### QUESTION

Let  $A = \{1, 2, 3, 4, 5\}$  and  $B = \{3, 4, 5, 6, 7\}$ . Perform the following operations: (i)  $A \cup B$  (ii)  $A \cap B$   
(iii)  $A - B$  (iv)  $B - A$ . [8]

##### ANSWER

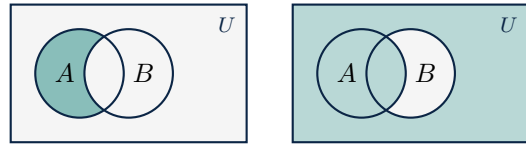
(i)  $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$  (ii)  $A \cap B = \{3, 4, 5\}$  (iii)  $A - B = \{1, 2\}$  (iv)  $B - A = \{6, 7\}$

## [SEC Final Examination 2024, CSE 0541 1143] Venn Diagram Shading

## QUESTION

Consider Universal set  $U$  and two sets  $A$  and  $B$ . Shade the following sets using a Venn diagram:  
 (i)  $A \cap B^C$  (ii)  $(B/A)^C$ . [4]

## ANSWER



(i)  $A \cap \overline{B} = A \setminus B$  (ii)  $\overline{B \setminus A} = \overline{B} \cup A$

(i) By the difference law,  $A \cap \overline{B} = A \setminus B$ : shaded strictly inside  $A$ , excluding  $A \cap B$ . (ii) Complement of the crescent  $B \setminus A$ : shaded throughout  $U$  except that crescent.

## 6 The Algebra of Sets & De Morgan's Laws

Because set operations are isomorphic to Boolean propositional logic ( $\cup \leftrightarrow \vee$ ,  $\cap \leftrightarrow \wedge$ ,  $\overline{A} \leftrightarrow \neg p$ ), set algebra obeys an identical set of algebraic laws:

| Law Name          | Union ( $\cup$ ) Form                                  | Intersection ( $\cap$ ) Form                           |
|-------------------|--------------------------------------------------------|--------------------------------------------------------|
| Identity Laws     | $A \cup \emptyset = A$                                 | $A \cap U = A$                                         |
| Domination Laws   | $A \cup U = U$                                         | $A \cap \emptyset = \emptyset$                         |
| Idempotent Laws   | $A \cup A = A$                                         | $A \cap A = A$                                         |
| Double Complement |                                                        | $\overline{\overline{A}} = A$                          |
| Commutative Laws  | $A \cup B = B \cup A$                                  | $A \cap B = B \cap A$                                  |
| Associative Laws  | $(A \cup B) \cup C = A \cup (B \cup C)$                | $(A \cap B) \cap C = A \cap (B \cap C)$                |
| Distributive Laws | $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$       | $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$       |
| De Morgan's Laws  | $\overline{A \cup B} = \overline{A} \cap \overline{B}$ | $\overline{A \cap B} = \overline{A} \cup \overline{B}$ |
| Absorption Laws   | $A \cup (A \cap B) = A$                                | $A \cap (A \cup B) = A$                                |
| Complement Laws   | $A \cup \overline{A} = U$                              | $A \cap \overline{A} = \emptyset$                      |

### Rigorous Analytical Proof: First De Morgan Law

**Theorem:** Prove that  $\overline{A \cup B} = \overline{A} \cap \overline{B}$ .

**Proof Strategy (Mutual Subset Inclusion):** To prove two sets  $X = Y$ , we prove  $X \subseteq Y$  and  $Y \subseteq X$ . Equivalently, we use chains of logical equivalences:

$$\begin{aligned}
 x \in \overline{A \cup B} &\iff x \notin (A \cup B) && \text{(Definition of complement)} \\
 &\iff \neg(x \in (A \cup B)) && \text{(Definition of } \notin) \\
 &\iff \neg(x \in A \vee x \in B) && \text{(Definition of union)} \\
 &\iff (\neg(x \in A)) \wedge (\neg(x \in B)) && \text{(De Morgan's Law of Propositional Logic!)} \\
 &\iff (x \notin A) \wedge (x \notin B) && \text{(Definition of } \notin) \\
 &\iff (x \in \overline{A}) \wedge (x \in \overline{B}) && \text{(Definition of complement)} \\
 &\iff x \in (\overline{A} \cap \overline{B}) && \text{(Definition of intersection)}
 \end{aligned}$$

Since each step is a bidirectional logical equivalence ( $\iff$ ), it follows that:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}. \quad \blacksquare$$

### [SEC Final Examination 2024, CSE 0541 1143] Set Identity Proofs & Absorption

#### QUESTION

Prove the following using set identities and logical equivalences:

- (i)  $A \cap (A \cap B) = A \cap B$  (*Printed in paper; also proved for  $A \cup (A \cap B) = A$* )  
 (ii)  $A \cap (A \cup B) = A$  (*Absorption Law*)

[05]

#### ANSWER

**Proof of (i):** By the Associative Law of set intersection:

$$A \cap (A \cap B) = (A \cap A) \cap B$$

By the Idempotent Law ( $A \cap A = A$ ):

$$(A \cap A) \cap B = A \cap B. \quad \blacksquare$$

*Note on Examination Variant ( $A \cup (A \cap B) = A$ ):*

$$\begin{aligned} x \in A \cup (A \cap B) &\iff x \in A \vee (x \in A \wedge x \in B) \\ &\iff (x \in A \vee x \in A) \wedge (x \in A \vee x \in B) && \text{(Distributivity)} \\ &\iff x \in A \wedge (x \in A \vee x \in B) \iff x \in A. && \text{(Absorption in Logic)} \end{aligned}$$

**Proof of (ii)  $A \cap (A \cup B) = A$  (Absorption Law):** We establish equality via mutual subset inclusion:

- Show  $A \cap (A \cup B) \subseteq A$ :** Let  $x \in A \cap (A \cup B)$ . By definition of intersection,  $x \in A$  and  $x \in (A \cup B)$ . In particular,  $x \in A$ . Therefore,  $A \cap (A \cup B) \subseteq A$ .
- Show  $A \subseteq A \cap (A \cup B)$ :** Let  $x \in A$ . Then certainly  $x \in A \cup B$  (by definition of union). Since  $x \in A$  and  $x \in A \cup B$ , it follows that  $x \in A \cap (A \cup B)$ . Therefore,  $A \subseteq A \cap (A \cup B)$ .

Since both inclusions hold,  $A \cap (A \cup B) = A$ .  $\blacksquare$

## 7 The Principle of Inclusion-Exclusion (PIE)

When counting elements in overlapping sets, simply adding  $|A| + |B|$  double-counts the common intersection. We must subtract the overlap to restore balance:

### 7.1 PIE for 2 Sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

### [SEC TT-01, CSE 103, 2022] Two-Set Overlap: Class Attendance Lists

#### QUESTION

List  $A$  contains the 30 students in a mathematics class, and list  $B$  contains the 35 students in an English class; 20 names appear on both lists. Find the number of students: (a) only on list  $A$ , (b) only on list  $B$ , (c) on list  $A$  or  $B$  (or both), (d) on exactly one list.

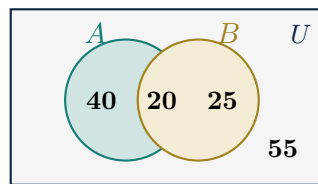
[5]

**ANSWER**

$$|A \setminus B| = 30 - 20 = 10 \quad |B \setminus A| = 35 - 20 = 15 \quad |A \cup B| = 30 + 35 - 20 = 45 \quad |A \oplus B| = 10 + 15 = 25$$

**[SEC Final Examination 2022, CSE 103] Venn Diagram Survey of Requirements****QUESTION**

Each student in Liberal Arts has a mathematics requirement  $A$  and a science requirement  $B$ . A poll of 140 sophomore students shows: 60 completed  $A$ , 45 completed  $B$ , 20 completed both. Use a Venn diagram to find the number of students who completed: (i) at least one of  $A$  and  $B$ ; (ii) exactly one of  $A$  or  $B$ ; (iii) neither  $A$  nor  $B$ . [1×3]

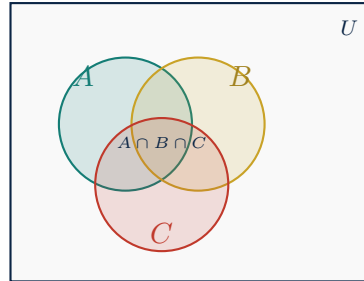
**ANSWER**

- (i)  $|A \cup B| = 60 + 45 - 20 = 85$ .    (ii)  $|A \setminus B| + |B \setminus A| = 40 + 25 = 65$ .    (iii)  $|U| - |A \cup B| = 140 - 85 = 55$ .

## 7.2 PIE for 3 Sets

For three arbitrary finite sets  $A$ ,  $B$ , and  $C$ :

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$



**Figure 3:** The 3-set overlapping geometry of the Inclusion-Exclusion Principle.

### [SEC Final Examination 2024, CSE 0541 1143] Three-Set Language Enrollment

#### QUESTION

In a class of 80 students, 50 know English ( $E$ ), 55 know French ( $F$ ), 46 know German ( $G$ ). 37 know English and French, 28 know French and German, 21 know English and German. 7 know none of the three languages. How many students know all three languages? [04]

#### ANSWER

Knowing at least one:  $|E \cup F \cup G| = 80 - 7 = 73$ . By PIE:

$$73 = 50 + 55 + 46 - (37 + 28 + 21) + |E \cap F \cap G| = 65 + |E \cap F \cap G| \implies |E \cap F \cap G| = 8$$

**8 students** know all three languages.

### [SEC Final Examination 2024, CSE 0541 1143] Three-Set Subject Preferences

#### QUESTION

Out of 100 students, 50 like English ( $E$ ), 35 like Science ( $S$ ), 29 like Math ( $M$ ). 6 like Math and English, 8 like English and Science, 6 like Science and Math. Only 5 like all three subjects. How many students don't like any of these subjects? (Use Venn diagram) [04]

#### ANSWER

$$|E \cup S \cup M| = 50 + 35 + 29 - (8 + 6 + 6) + 5 = 114 - 20 + 5 = 99$$

Students liking none:  $|U| - |E \cup S \cup M| = 100 - 99 = 1$  student.

## 8 ★ Functions: Definitions, Types, Inverses & Compositions

★ **SEC Syllabus Core Topic:** *Functions, Domain/Codomain/Range, Injections, Surjections, Bijections, Inverses, Compositions, Floor/Ceiling*

A **function** (or *mapping*) is a foundational mathematical construct underpinning functional programming, algorithmic analysis, hashing, and cryptography. Set-theoretically, a function is a constrained binary relation.

### 8.1 Formal Definition of a Function

#### Formal Definition of a Function

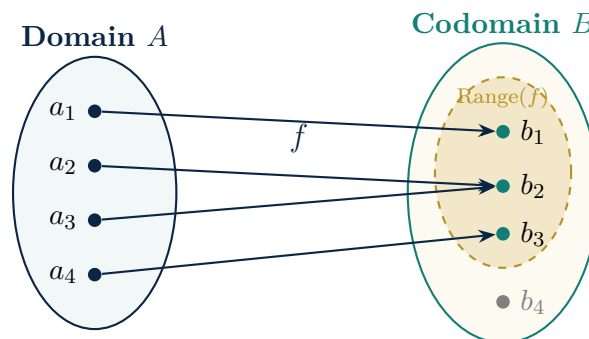
Let  $A$  and  $B$  be non-empty sets. A **function**  $f$  from  $A$  to  $B$ , denoted:

$$f : A \rightarrow B$$

is a relation  $f \subseteq A \times B$  that assigns to **each** element  $a \in A$  **exactly one** element  $b \in B$ . We write  $f(a) = b$ , where:

- $A$  is the **Domain** of  $f$ , denoted  $\text{Dom}(f) = A$ .
- $B$  is the **Codomain** of  $f$ , denoted  $\text{Codom}(f) = B$ .
- $b$  is the **image** of  $a$  under  $f$ , and  $a$  is a **preimage** of  $b$ .
- The **Range** (or *Image of  $A$* ) is the set of all actual outputs:

$$\text{Range}(f) = f(A) = \{f(a) \mid a \in A\} = \{b \in B \mid \exists a \in A \text{ such that } f(a) = b\}$$



**Figure 4:** Anatomy of a function: Domain  $A$ , Codomain  $B$ , and Range  $f(A) \subseteq B$ . Note  $b_4 \notin \text{Range}(f)$ .

#### The Universal Beginner Trap: Codomain vs. Range

Never confuse the **Codomain** with the **Range**!

- The **Codomain** is the target set specified in the function declaration ( $B$  in  $f : A \rightarrow B$ ).
- The **Range** is the set of elements in  $B$  that are actually hit by at least one  $a \in A$ .
- In general,  $\text{Range}(f) \subseteq \text{Codom}(f)$ . They are equal **if and only if**  $f$  is **surjective** (onto).

#### Equality Criterion for Two Functions

Two functions  $f$  and  $g$  are **equal** ( $f = g$ ) if and only if:

1.  $\text{Dom}(f) = \text{Dom}(g)$ ,
2.  $\text{Codom}(f) = \text{Codom}(g)$ , and
3.  $f(x) = g(x)$  for all  $x \in \text{Dom}(f)$ .

*Example:*  $f : \mathbb{R} \rightarrow \mathbb{R}$  where  $f(x) = x$  and  $g : \mathbb{Z} \rightarrow \mathbb{R}$  where  $g(x) = x$  are **NOT equal** because  $\text{Dom}(f) \neq \text{Dom}(g)$ .

## 8.2 One-to-One Functions (Injections)

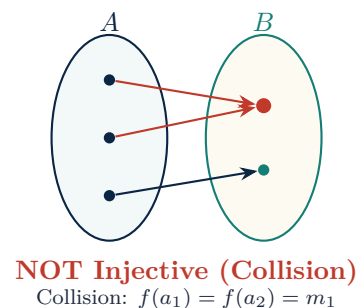
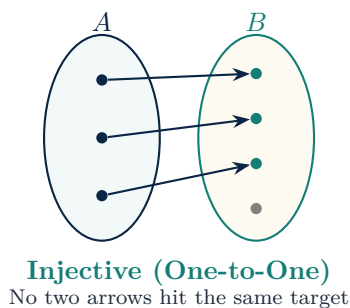
### Injective / One-to-One Function

A function  $f : A \rightarrow B$  is **one-to-one** (or an **injection**) if and only if distinct elements in the domain have distinct images in the codomain.

$$\forall x_1, x_2 \in A, \quad f(x_1) = f(x_2) \implies x_1 = x_2$$

Equivalently, by contraposition:

$$\forall x_1, x_2 \in A, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$



### Analytical Proofs: Proving vs. Disproving Injectivity

1. **Prove**  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = 4x - 7$  is **Injective**:

*Proof:* Assume  $f(x_1) = f(x_2)$ . Then:

$$4x_1 - 7 = 4x_2 - 7 \implies 4x_1 = 4x_2 \implies x_1 = x_2.$$

Since  $f(x_1) = f(x_2) \implies x_1 = x_2$ ,  $f$  is strictly injective. ■

2. **Disprove that**  $g : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $g(x) = x^2$  is **Injective**:

*Refutation (Counter-example Method):* To disprove  $\forall x_1, x_2 (g(x_1) = g(x_2) \implies x_1 = x_2)$ , find one explicit pair where  $x_1 \neq x_2$  but  $g(x_1) = g(x_2)$ .

Choose  $x_1 = 2$  and  $x_2 = -2$ . Notice  $x_1 \neq x_2$ , but:

$$g(2) = 2^2 = 4 \quad \text{and} \quad g(-2) = (-2)^2 = 4 \implies g(2) = g(-2).$$

Therefore,  $g$  is **not injective**.

## 8.3 Onto Functions (Surjections)

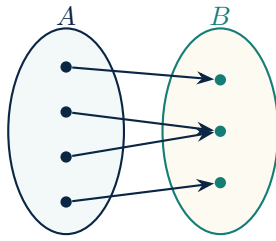
### Surjective / Onto Function

A function  $f : A \rightarrow B$  is **onto** (or a **surjection**) if and only if every element in the codomain  $B$  is the image of at least one element in the domain  $A$ :

$$\forall y \in B, \quad \exists x \in A \quad \text{such that} \quad f(x) = y$$

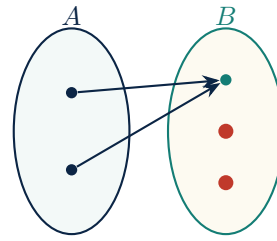
Equivalently:

$$\text{Range}(\mathbf{f}) = \text{Codomain}(\mathbf{f}) = \mathbf{B}$$



**Surjective (Onto)**

Every element in  $B$  has incoming arrow



**NOT Surjective**

Uncovered:  $v_2, v_3 \in B$  have no preimage

### Analytical Proofs: Proving vs. Disproving Surjectivity

#### 1. Prove $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 3x - 5$ is Surjective:

*Proof Strategy (Finding Preimage):* Let  $y \in \mathbb{R}$  be an arbitrary element in the codomain. We must find  $x \in \mathbb{R}$  such that  $f(x) = y$ :

$$y = 3x - 5 \iff 3x = y + 5 \iff x = \frac{y + 5}{3}$$

Since  $y \in \mathbb{R}$ ,  $x = \frac{y+5}{3}$  is a well-defined real number ( $x \in \mathbb{R}$ ). Checking:

$$f\left(\frac{y+5}{3}\right) = 3\left(\frac{y+5}{3}\right) - 5 = (y+5) - 5 = y.$$

Hence, every  $y \in \mathbb{R}$  has a valid preimage in  $\mathbb{R}$ .  $f$  is surjective. ■

#### 2. Disprove that $g : \mathbb{Z} \rightarrow \mathbb{Z}$ with $g(x) = 2x$ is Surjective:

*Refutation:* The codomain is  $\mathbb{Z}$ . Consider  $y = 3 \in \mathbb{Z}$ . If  $g(x) = 3$ , then  $2x = 3 \implies x = 1.5 \notin \mathbb{Z}$ . Since the odd integer 3 has no preimage in the domain  $\mathbb{Z}$ ,  $g$  is **not surjective**.

## 8.4 One-to-One Correspondences (Bijections)

### Bijjective Function / One-to-One Correspondence

A function  $f : A \rightarrow B$  is a **bijection** (or a **one-to-one correspondence**) if and only if it is **both injective and surjective**.

$$\mathbf{f} \text{ is Bijjective} \iff (\mathbf{f} \text{ is Injective} \wedge \mathbf{f} \text{ is Surjective})$$

| Classification           | Geometric / Relational Meaning                  | Cardinality (Finite Sets) | Constraint |
|--------------------------|-------------------------------------------------|---------------------------|------------|
| <b>Injective Only</b>    | Each $b \in B$ receives at most 1 arrow.        | $ A  \leq  B $            |            |
| <b>Surjective Only</b>   | Each $b \in B$ receives at least 1 arrow.       | $ A  \geq  B $            |            |
| <b>Bijjective (Both)</b> | Each $b \in B$ receives <b>exactly 1</b> arrow. | $ A  =  B $               |            |
| <b>Neither</b>           | Collisions exist AND some $b \in B$ are missed. | No cardinality constraint |            |

## 8.5 Inverse Functions ( $f^{-1}$ )

### Inverse Function Definition

Let  $f : A \rightarrow B$  be a **bijective** function. The **inverse function** of  $f$ , denoted  $f^{-1} : B \rightarrow A$ , assigns to each element  $b \in B$  the unique element  $a \in A$  such that  $f(a) = b$ :

$$f^{-1}(b) = a \iff f(a) = b$$

### Invertibility Criterion Theorem

A function  $f : A \rightarrow B$  is **invertible** if and only if it is a **bijection**.

- If  $f$  is *not injective*, two distinct domain elements  $a_1 \neq a_2$  map to the same  $b$ . Then  $f^{-1}(b)$  would have to return multiple values ( $\{a_1, a_2\}$ ), violating the single-valued definition of a function.
- If  $f$  is *not surjective*, some  $b \in B$  has no preimage. Then  $f^{-1}(b)$  would be undefined, violating the requirement that a function must map every element of its domain.

### Notation Alert: Inverse Function vs. Reciprocal

Do not confuse  $f^{-1}(x)$  with the algebraic reciprocal  $\frac{1}{f(x)}$ !

$$f^{-1}(x) \neq (f(x))^{-1} = \frac{1}{f(x)}$$

$f^{-1}$  denotes the **functional inverse**, whereas  $\frac{1}{f(x)}$  is the **multiplicative inverse** of the numerical output.

### Algorithmic Procedure to Find an Inverse Function

**Task:** Let  $f : \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R} \setminus \{1\}$  defined by  $f(x) = \frac{x+3}{x-2}$ . Find  $f^{-1}(x)$ .

1. **Step 1: Set  $y = f(x)$ :**

$$y = \frac{x+3}{x-2}$$

2. **Step 2: Solve algebraically for  $x$  in terms of  $y$ :**

$$y(x-2) = x+3 \implies xy - 2y = x+3 \implies xy - x = 2y+3 \implies x(y-1) = 2y+3$$

$$x = \frac{2y+3}{y-1} \quad (\text{defined since } y \neq 1)$$

3. **Step 3: Interchange variables to express  $f^{-1}(x)$ :**

$$f^{-1}(x) = \frac{2x+3}{x-1}$$

4. **Step 4: Verification ( $f(f^{-1}(x)) = x$ ):**

$$f(f^{-1}(x)) = \frac{\frac{2x+3}{x-1} + 3}{\frac{2x+3}{x-1} - 2} = \frac{\frac{2x+3+3x-3}{x-1}}{\frac{2x+3-2x+2}{x-1}} = \frac{5x}{5} = x. \quad \checkmark$$

## 8.6 Composition of Functions ( $f \circ g$ )

### Composition of Functions

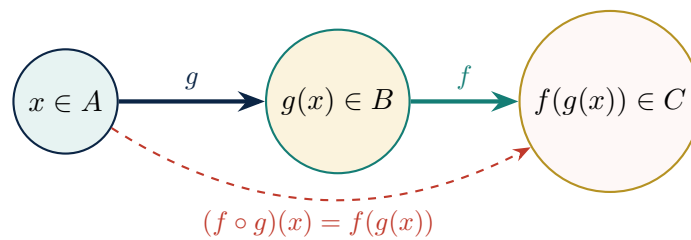
Let  $g : A \rightarrow B$  and  $f : B \rightarrow C$ . The **composition** of  $f$  and  $g$ , denoted by:

$$f \circ g : A \rightarrow C$$

is defined for all  $x \in A$  by:

$$(f \circ g)(x) = f(g(x))$$

*Note the evaluation order:*  $g$  operates first on  $x$ , and then  $f$  operates on the resulting value  $g(x)$ .



### Algebraic Properties of Composition

1. **Associativity:** If  $h : A \rightarrow B$ ,  $g : B \rightarrow C$ , and  $f : C \rightarrow D$ , then:

$$f \circ (g \circ h) = (f \circ g) \circ h$$

2. **Non-Commutativity (Crucial!):** In general:

$$f \circ g \neq g \circ f$$

*Counterexample:* Let  $f(x) = 2x$  and  $g(x) = x + 1$ .

$$(f \circ g)(x) = f(g(x)) = 2(x + 1) = 2x + 2, \quad (g \circ f)(x) = g(f(x)) = 2x + 1 \implies f \circ g \neq g \circ f.$$

3. **Identity with Inverse:** If  $f : A \rightarrow B$  is a bijection, then:

$$f^{-1} \circ f = \text{id}_A \quad \text{and} \quad f \circ f^{-1} = \text{id}_B$$

4. **Preservation of Injectivity / Surjectivity / Bijectivity:**

- If  $f$  and  $g$  are both injective, then  $f \circ g$  is **injective**.
- If  $f$  and  $g$  are both surjective, then  $f \circ g$  is **surjective**.
- If  $f$  and  $g$  are both bijective, then  $f \circ g$  is **bijective**, and  $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$ .

## 8.7 Essential Special Functions in Computer Science

### Floor and Ceiling Functions

- The **Floor function**  $\lfloor x \rfloor$  assigns to the real number  $x$  the largest integer that is less than or equal to  $x$ :

$$\lfloor x \rfloor = n \iff n \leq x < n + 1, \quad n \in \mathbb{Z}$$

- The **Ceiling function**  $\lceil x \rceil$  assigns to the real number  $x$  the smallest integer that is greater than or equal to  $x$ :

$$\lceil x \rceil = n \iff n - 1 < x \leq n, \quad n \in \mathbb{Z}$$

## Useful Mathematical Properties of Floor and Ceiling

| Property Name              | Mathematical Formula                                                                             | Context ( $n \in \mathbb{Z}, x \in \mathbb{R}$ ) |
|----------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Integer Identity</b>    | $\lfloor x \rfloor = x \iff x \in \mathbb{Z} \iff \lceil x \rceil = x$                           | Exact integer equality                           |
| <b>Bounds</b>              | $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$                                  | Tight squeeze inequalities                       |
| <b>Integer Shift</b>       | $\lfloor x + n \rfloor = \lfloor x \rfloor + n, \quad \lceil x + n \rceil = \lceil x \rceil + n$ | Pulling integers outside                         |
| <b>Negative Symmetry</b>   | $\lfloor -x \rfloor = -\lceil x \rceil, \quad \lceil -x \rceil = -\lfloor x \rfloor$             | Sign flipping inversion                          |
| <b>Duplication Formula</b> | $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$                       | Hermite's Identity ( $k = 2$ )                   |

## CS Algorithmic Applications of Floor &amp; Ceiling

- **Binary Bit-Length of an Integer  $n$ :** The number of bits required to represent a positive integer  $n$  in binary is:

$$b(n) = \lfloor \log_2 n \rfloor + 1$$

*Example:* For  $n = 25$ ,  $\log_2(25) \approx 4.6438 \implies \lfloor 4.6438 \rfloor + 1 = 4 + 1 = \mathbf{5}$  bits ( $25_{10} = 11001_2$ ).

- **Height of a Complete Binary Tree:** A complete binary tree with  $n$  nodes has height:

$$h = \lceil \log_2(n + 1) \rceil - 1$$

- **Page Allocation in Memory / Pagination:** Dividing  $N$  data records into pages of size  $S$  requires:

$$\text{Pages} = \left\lceil \frac{N}{S} \right\rceil$$

## 8.8 Past SEC Examination Questions on Functions

## [SEC Final Examination, CSE 0541 1143] Injectivity, Surjectivity &amp; Inverse

## QUESTION

Let  $A = \mathbb{R} \setminus \{2\}$  and  $B = \mathbb{R} \setminus \{1\}$ . Define  $f : A \rightarrow B$  by:

$$f(x) = \frac{x-3}{x-2}$$

- (a) Prove that  $f$  is bijective. [4]  
 (b) Find the explicit formula for  $f^{-1}(x)$  and state its domain. [2]

## ANSWER

## (a) Proof of Bijectivity:

1. **Injectivity:** Let  $x_1, x_2 \in A$  such that  $f(x_1) = f(x_2)$ :

$$\frac{x_1 - 3}{x_1 - 2} = \frac{x_2 - 3}{x_2 - 2} \implies (x_1 - 3)(x_2 - 2) = (x_2 - 3)(x_1 - 2)$$

Expanding both sides:

$$x_1x_2 - 2x_1 - 3x_2 + 6 = x_1x_2 - 2x_2 - 3x_1 + 6$$

Canceling  $x_1x_2 + 6$  from both sides:

$$-2x_1 - 3x_2 = -2x_2 - 3x_1 \implies 3x_1 - 2x_1 = 3x_2 - 2x_2 \implies x_1 = x_2.$$

Thus,  $f$  is strictly **injective**.

2. **Surjectivity:** Let  $y \in B = \mathbb{R} \setminus \{1\}$ . We show there exists  $x \in A$  such that  $f(x) = y$ :

$$y = \frac{x-3}{x-2} \implies y(x-2) = x-3 \implies xy - 2y = x-3 \implies x(y-1) = 2y-3$$

Since  $y \neq 1$ , division by  $(y-1)$  is valid:

$$x = \frac{2y-3}{y-1}$$

Is  $x \in A$ ? Notice  $x = 2 \iff \frac{2y-3}{y-1} = 2 \iff 2y-3 = 2y-2 \iff -3 = -2$  (impossible).

Hence  $x \neq 2$ , so  $x \in A$ . Thus, every  $y \in B$  has a valid preimage in  $A$ .  $f$  is **surjective**.

Since  $f$  is both injective and surjective,  $f$  is a **bijection**. ■

(b) **Finding  $f^{-1}(x)$ :** From the preimage equation derived in (a), replacing  $y$  with  $x$ :

$$f^{-1}(x) = \frac{2x-3}{x-1}, \quad \text{Dom}(f^{-1}) = \mathbb{R} \setminus \{1\}.$$

### [SEC Term Final Exam, CSE 103] Proof of Hermite's Floor Identity

#### QUESTION

Prove that for any real number  $x \in \mathbb{R}$ :

$$\lfloor 2x \rfloor = \lfloor x \rfloor + \left\lfloor x + \frac{1}{2} \right\rfloor$$

[4]

#### ANSWER

Any real number  $x$  can be decomposed into its integer part and fractional part:

$$x = n + \epsilon, \quad \text{where } n = \lfloor x \rfloor \in \mathbb{Z} \quad \text{and } 0 \leq \epsilon < 1.$$

We divide the fractional part  $\epsilon$  into two exhaustive cases:

- **Case 1:**  $0 \leq \epsilon < \frac{1}{2}$ :
  - $2x = 2(n + \epsilon) = 2n + 2\epsilon$ . Since  $0 \leq 2\epsilon < 1$ , we have  $\lfloor 2x \rfloor = 2n$ .
  - $x + \frac{1}{2} = n + \epsilon + \frac{1}{2}$ . Since  $0 \leq \epsilon < \frac{1}{2}$ , we have  $\frac{1}{2} \leq \epsilon + \frac{1}{2} < 1$ , so  $\lfloor x + \frac{1}{2} \rfloor = n$ .
  - RHS:  $\lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor = n + n = 2n = \text{LHS}$ . Holds!
- **Case 2:**  $\frac{1}{2} \leq \epsilon < 1$ :
  - $2x = 2n + 2\epsilon = 2n + 1 + (2\epsilon - 1)$ . Since  $0 \leq 2\epsilon - 1 < 1$ , we have  $\lfloor 2x \rfloor = 2n + 1$ .
  - $x + \frac{1}{2} = n + (\epsilon + \frac{1}{2})$ . Since  $1 \leq \epsilon + \frac{1}{2} < \frac{3}{2}$ , we have  $\lfloor x + \frac{1}{2} \rfloor = n + 1$ .
  - RHS:  $\lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor = n + (n + 1) = 2n + 1 = \text{LHS}$ . Holds!

Since the equality holds in both exhaustive cases,  $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$  for all  $x \in \mathbb{R}$ . ■

### [SEC Final Examination 2022, CSE 103] Step Function Graphs: $\lfloor x/2 \rfloor$ and $\lceil x/2 \rceil$

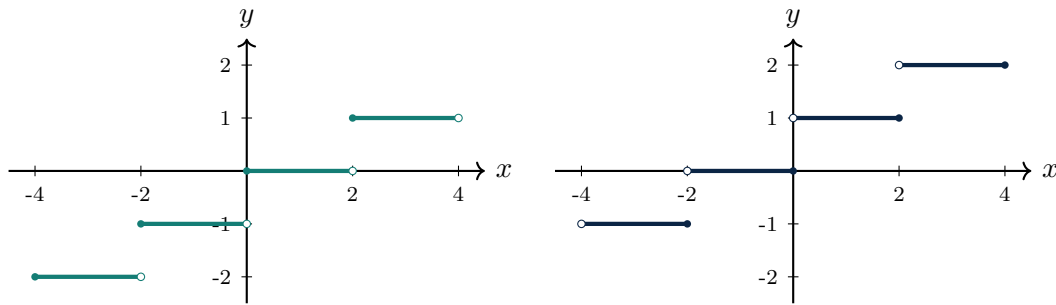
#### QUESTION

Plot the following functions over the interval  $[-4, 4]$ : (i)  $\left\lceil \frac{x}{2} \right\rceil$  (ii)  $\left\lfloor \frac{x}{2} \right\rfloor$  [4]

#### ANSWER

Both functions are piecewise constant step functions with step discontinuities at every even integer

$(x \in 2\mathbb{Z})$ :



(a)  $y = \lfloor \frac{x}{2} \rfloor$  (left-closed, right-open)

(b)  $y = \lceil \frac{x}{2} \rceil$  (left-open, right-closed)

- For  $y = \lfloor x/2 \rfloor$ : On  $[-4, -2)$ ,  $y = -2$ ; on  $[-2, 0)$ ,  $y = -1$ ; on  $[0, 2)$ ,  $y = 0$ ; on  $[2, 4)$ ,  $y = 1$ .
- For  $y = \lceil x/2 \rceil$ : On  $(-4, -2]$ ,  $y = -1$ ; on  $(-2, 0]$ ,  $y = 0$ ; on  $(0, 2]$ ,  $y = 1$ ; on  $(2, 4]$ ,  $y = 2$ .

### [SEC Final Examination 2022, CSE 103] Multi-Stage Mapping Anatomy (Figure-1)

#### QUESTION

Let the functions  $f: A \rightarrow B$ ,  $g: B \rightarrow C$ , and  $h: C \rightarrow D$  be defined by Figure-1 below. Determine if each function is: (i) onto, (ii) one-to-one, (iii) invertible. [1×3]

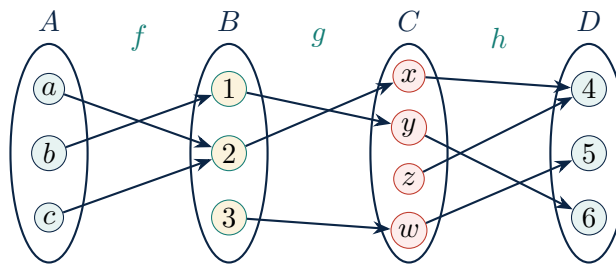


Figure-1: Functional chains across four finite alphabets.

#### ANSWER

A function is **invertible** if and only if it is a **bijection** (both one-to-one and onto):

##### 1. Function $f: A \rightarrow B$ :

- **One-to-one: No** — both  $a$  and  $c$  map to the same image:  $f(a) = f(c) = 2$  ( $a \neq c$ ).
- **Onto: No** — element  $3 \in B$  has no preimage;  $\text{Range}(f) = \{1, 2\} \neq B$ .
- **Invertible: No** (not a bijection).

##### 2. Function $g: B \rightarrow C$ :

- **One-to-one: Yes** — distinct inputs produce distinct outputs:  $g(1) = y, g(2) = x, g(3) = w$ .
- **Onto: No** — target element  $z \in C$  has no incoming edge;  $\text{Range}(g) = \{x, y, w\} \subsetneq C$ .
- **Invertible: No** (fails surjectivity).

##### 3. Function $h: C \rightarrow D$ :

- **One-to-one: No** — collision at output 4:  $h(x) = h(z) = 4$  ( $x \neq z$ ).
- **Onto: Yes** — every target element in  $D = \{4, 5, 6\}$  has at least one preimage:  $h(x) = 4, h(w) = 5, h(y) = 6$ .
- **Invertible: No** (fails injectivity).

## [SEC Final 2022 &amp; CSE 143 Final 2023] Floor &amp; Ceiling Numerical Evaluations

## QUESTION

Evaluate the exact numerical value of each expression:

(a)  $\lfloor 7.5 \rfloor, \lfloor -7.5 \rfloor, \lfloor -18 \rfloor$       (b)  $\lceil 7.5 \rceil, \lceil -7.5 \rceil, \lceil -18 \rceil$       (c)  $\left\lfloor \frac{1}{5} \right\rfloor, \lceil 2.5 \rceil, \lfloor -2.99 \rfloor, \left\lceil -\frac{1}{2} \right\rceil$

$[2+2]$

## ANSWER

Recall that  $\lfloor x \rfloor$  is the largest integer  $\leq x$ , and  $\lceil x \rceil$  is the smallest integer  $\geq x$ :

- $\lfloor 7.5 \rfloor = 7$ ,  $\lfloor -7.5 \rfloor = -8$  (since  $-8 < -7.5 < -7$ ),  $\lfloor -18 \rfloor = -18$ .
- $\lceil 7.5 \rceil = 8$ ,  $\lceil -7.5 \rceil = -7$ ,  $\lceil -18 \rceil = -18$ .
- $\left\lfloor \frac{1}{5} \right\rfloor = \lfloor 0.2 \rfloor = 0$ .
- $\lceil 2.5 \rceil = 3$ .
- $\lfloor -2.99 \rfloor = -3$  (the greatest integer  $\leq -2.99$  is  $-3$ ).
- $\left\lceil -\frac{1}{2} \right\rceil = \lceil -0.5 \rceil = 0$ .

[SEC Final 2022 / TT-01 / Final 2024] The Ackermann Function:  $A(1,3)$  and  $A(2,2)$ 

## QUESTION

The **Ackermann function**  $A(m, n)$  on non-negative integers  $\mathbb{N}_0 \times \mathbb{N}_0$  is defined recursively by:

$$A(m, n) = \begin{cases} n + 1 & \text{if } m = 0 \\ A(m - 1, 1) & \text{if } m > 0 \text{ and } n = 0 \\ A(m - 1, A(m, n - 1)) & \text{if } m > 0 \text{ and } n > 0 \end{cases}$$

- (a) State the conditions and compute  $A(1, 3)$ .  $[5]$   
 (b) Use the formal definition to evaluate  $A(2, 2)$ .  $[3]$

## ANSWER

**(a) Step-by-step Evaluation of  $A(1, 3)$ :** First, evaluate the base sequence for  $m = 1$ :

$$\begin{aligned} A(1, 0) &= A(0, 1) = 1 + 1 = 2 \\ A(1, 1) &= A(0, A(1, 0)) = A(0, 2) = 2 + 1 = 3 \\ A(1, 2) &= A(0, A(1, 1)) = A(0, 3) = 3 + 1 = 4 \\ A(1, 3) &= A(0, A(1, 2)) = A(0, 4) = 4 + 1 = 5 \end{aligned}$$

*Closed-form law for  $m = 1$ :* In general,  $A(1, n) = n + 2$ . Thus,  $A(1, 3) = 5$ .

**(b) Step-by-step Evaluation of  $A(2, 2)$ :** For  $m = 2$ , we apply  $A(2, n) = A(1, A(2, n - 1))$  and use our closed form  $A(1, k) = k + 2$ :

$$\begin{aligned} A(2, 0) &= A(1, 1) = 1 + 2 = 3 \\ A(2, 1) &= A(1, A(2, 0)) = A(1, 3) = 3 + 2 = 5 \\ A(2, 2) &= A(1, A(2, 1)) = A(1, 5) = 5 + 2 = 7 \end{aligned}$$

*Closed-form law for  $m = 2$ :* In general,  $A(2, n) = 2n + 3$ . Thus,  $A(2, 2) = 7$ .

[SEC Final Examination 2023, CSE 143] Bijection Analysis of  $f(x) = x^2 + 1$  on  $\mathbb{R} \rightarrow \mathbb{R}$ 

## QUESTION

Let  $f(x) = x^2 + 1$  be a function from  $\mathbb{R}$  to  $\mathbb{R}$ . Is it a one-to-one correspondence (bijection)? Justify your answer. [4]

## ANSWER

A one-to-one correspondence requires the function to be simultaneously **injective** and **surjective**:

1. **Injectivity (One-to-One) Test:** Consider  $x_1 = 1$  and  $x_2 = -1$ :

$$f(1) = 1^2 + 1 = 2, \quad f(-1) = (-1)^2 + 1 = 2.$$

Since  $f(1) = f(-1)$  but  $1 \neq -1$ ,  $f$  fails the injectivity criterion.  $f$  is **NOT one-to-one**.

2. **Surjectivity (Onto) Test:** For any real number  $x \in \mathbb{R}$ ,  $x^2 \geq 0 \implies x^2 + 1 \geq 1$ . Thus, the range of  $f$  is:

$$\text{Range}(f) = [1, \infty) \subsetneq \mathbb{R}.$$

Any target element  $y < 1$  (for example,  $y = 0 \in \mathbb{R}$ ) has no real preimage because  $x^2 + 1 = 0 \implies x^2 = -1$  has no solution in  $\mathbb{R}$ . Thus,  $f$  is **NOT onto**.

**Conclusion:** Because  $f$  fails both injectivity and surjectivity, it is **NOT a one-to-one correspondence**.

## [SEC Final Examination 2023, CSE 143] Integer Function Compositions

## QUESTION

Let  $f$  and  $g$  be functions from  $\mathbb{Z}$  to  $\mathbb{Z}$  defined by  $f(x) = 2x + 3$  and  $g(x) = 3x^2 + 2$ . Determine  $f \circ g(3)$  and  $g \circ f(1)$ . [4]

## ANSWER

By definition of function composition,  $(f \circ g)(x) = f(g(x))$  and  $(g \circ f)(x) = g(f(x))$ :

1. **Compute  $f \circ g(3)$ :** First evaluate the inner function at  $x = 3$ :

$$g(3) = 3(3^2) + 2 = 3(9) + 2 = 27 + 2 = 29$$

Now evaluate the outer function at 29:

$$f(g(3)) = f(29) = 2(29) + 3 = 58 + 3 = \mathbf{61}.$$

2. **Compute  $g \circ f(1)$ :** First evaluate the inner function at  $x = 1$ :

$$f(1) = 2(1) + 3 = 2 + 3 = 5$$

Now evaluate the outer function at 5:

$$g(f(1)) = g(5) = 3(5^2) + 2 = 3(25) + 2 = 75 + 2 = \mathbf{77}.$$

Observe that  $f \circ g \neq g \circ f$ , verifying that function composition is non-commutative.

[SEC Final Examination, CSE 0541 1143] Natural Number Mapping:  $f(x) = 2x + 1$  on  $\mathbb{N} \rightarrow \mathbb{N}$ 

## QUESTION

A function  $f: \mathbb{N} \rightarrow \mathbb{N}$  is defined as  $f(x) = 2x + 1$  (where  $\mathbb{N} = \{1, 2, 3, \dots\}$ ). Determine if  $f$  is one-to-one and onto. Justify your answer. [03]

**ANSWER**

1. **One-to-One (Injective):** Let  $a, b \in \mathbb{N}$  such that  $f(a) = f(b)$ :

$$2a + 1 = 2b + 1 \implies 2a = 2b \implies a = b.$$

Since  $f(a) = f(b)$  implies  $a = b$  for all  $a, b \in \mathbb{N}$ ,  $f$  is strictly **one-to-one (injective)**.

2. **Onto (Surjective):** For  $f$  to be onto, every  $y \in \mathbb{N}$  must have a preimage  $x \in \mathbb{N}$  satisfying  $2x + 1 = y \implies x = \frac{y-1}{2}$ .

- For  $y = 2 \in \mathbb{N}$  (or any even number):  $x = \frac{2-1}{2} = \frac{1}{2} \notin \mathbb{N}$ .
- The range of  $f$  consists only of odd natural numbers:  $\text{Range}(f) = \{3, 5, 7, \dots\} \neq \mathbb{N}$ .

Since even natural numbers have no integer preimages in  $\mathbb{N}$ ,  $f$  is **NOT onto (not surjective)**.

[SEC Final Examination 2024, CSE 0541 1143] Triple Function Compositions on  $X = \{1, 2, 3\}$

**QUESTION**

Let  $f, g, h$  be functions on  $X = \{1, 2, 3\}$  defined as:

$$f = \{(1, 2), (2, 3), (3, 1)\}, \quad g = \{(1, 2), (2, 1), (3, 3)\}, \quad h = \{(1, 1), (2, 2), (3, 1)\}$$

Compute: (i)  $f \circ g \circ h$       (ii)  $f \circ h \circ g$ .

[04]

**ANSWER**

Recall that  $(f \circ g \circ h)(x) = f(g(h(x)))$  evaluates from **right to left**:

1. (i) **Compute  $f \circ g \circ h$ :**

- For  $x = 1$ :  $h(1) = 1 \implies g(1) = 2 \implies f(2) = 3 \implies (1, 3)$ .
  - For  $x = 2$ :  $h(2) = 2 \implies g(2) = 1 \implies f(1) = 2 \implies (2, 2)$ .
  - For  $x = 3$ :  $h(3) = 1 \implies g(1) = 2 \implies f(2) = 3 \implies (3, 3)$ .
- $$\mathbf{f \circ g \circ h = \{(1, 3), (2, 2), (3, 3)\}}$$

2. (ii) **Compute  $f \circ h \circ g$ :**

- For  $x = 1$ :  $g(1) = 2 \implies h(2) = 2 \implies f(2) = 3 \implies (1, 3)$ .
  - For  $x = 2$ :  $g(2) = 1 \implies h(1) = 1 \implies f(1) = 2 \implies (2, 2)$ .
  - For  $x = 3$ :  $g(3) = 3 \implies h(3) = 1 \implies f(1) = 2 \implies (3, 2)$ .
- $$\mathbf{f \circ h \circ g = \{(1, 3), (2, 2), (3, 2)\}}$$

[SEC TT-02 / Final Examination 2024] Non-Linear Recursive Sequence

**QUESTION**

Find  $f(4)$  if  $f$  is defined recursively on integers by:

$$f(0) = -1, \quad f(1) = 2, \quad \text{and} \quad f(n+1) = f(n)^2 \cdot f(n-1) \quad \text{for } n \geq 1.$$

[02 / 03]

**ANSWER**

We compute subsequent terms iteratively using the recursive relation:

1. For  $n = 1$ :

$$f(2) = f(1)^2 \cdot f(0) = (2)^2 \cdot (-1) = 4 \cdot (-1) = -4.$$

2. For  $n = 2$ :

$$f(3) = f(2)^2 \cdot f(1) = (-4)^2 \cdot (2) = 16 \cdot 2 = \mathbf{32}.$$

3. For  $n = 3$ :

$$f(4) = f(3)^2 \cdot f(2) = (32)^2 \cdot (-4) = 1024 \cdot (-4) = -4096.$$

Therefore,  $f(4) = -4096$ .

## 9 Summary

### 9.1 Fundamental Definitions, Axioms & Inclusion Statements

#### Core Axiomatic Statements & Definitions

- **Set Membership vs. Inclusion:**

$$x \in A \quad (\text{element belongs to set}) \quad \text{vs.} \quad A \subseteq B \iff (\forall x)(x \in A \implies x \in B)$$

- ★ **Set Equality Principle (Double Inclusion):**

$$A = B \iff (A \subseteq B \wedge B \subseteq A) \iff (\forall x)(x \in A \iff x \in B)$$

- ★ **The Empty Set ( $\emptyset$ ) Axioms:**

$$\emptyset = \{\}, \quad |\emptyset| = 0, \quad \emptyset \subseteq A \quad (\forall A), \quad \emptyset \in \mathcal{P}(A) \quad (\forall A)$$

- **Proper Subset Criterion:**

$$A \subset B \iff (A \subseteq B \wedge A \neq B)$$

- **Universal Set Bounds:**

$$\forall A \subseteq \mathcal{U}: \quad \emptyset \subseteq A \subseteq \mathcal{U}, \quad \overline{\mathcal{U}} = \emptyset, \quad \overline{\emptyset} = \mathcal{U}$$

- ★ **Fundamental Set Operations & Subsumption Equivalences:**

$$A \subseteq B \iff A \cap B = A \iff A \cup B = B \iff A \setminus B = \emptyset \iff \overline{B} \subseteq \overline{A}$$

- ★ **Binary Relations, Domain & Range:** For any relation  $R \subseteq A \times B$ :

$$\text{Dom}(R) = \{a \in A \mid \exists b \in B, (a, b) \in R\}, \quad \text{Range}(R) = \{b \in B \mid \exists a \in A, (a, b) \in R\}$$

- ★ **Reflexive Relation Criterion:**

$$\forall a \in A, (a, a) \in R \iff \text{All diagonal entries } m_{ii} = 1 \iff \text{Self-loops on all digraph vertices}$$

- ★ **Function Single-Valuedness Axiom:**

$$f : A \rightarrow B \iff \forall a \in A, \exists! b \in B, (a, b) \in f \quad \text{with } \text{Range}(f) \subseteq \text{Codom}(f)$$

- ★ **Invertibility Principle:**

$$f^{-1} : B \rightarrow A \text{ exists} \iff f \text{ is Bijective (Injective} \wedge \text{Surjective)}$$

- ★ **Composition Law & Identity:**

$$(f \circ g)(x) = f(g(x)), \quad f \circ f^{-1} = \text{id}_B, \quad f^{-1} \circ f = \text{id}_A$$

- **Floor & Ceiling Bound Axiom:**

$$x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1, \quad \lfloor x + n \rfloor = \lfloor x \rfloor + n \quad (\forall n \in \mathbb{Z})$$

## 9.2 Cardinality, Enumeration &amp; Combinatorial Formulae

| Essential Counting Formulae Compendium |                                                                            |                                          |
|----------------------------------------|----------------------------------------------------------------------------|------------------------------------------|
| Concept                                | Formula                                                                    | Condition / Context                      |
| ★ Set Cardinality                      | $ A  = n$                                                                  | Number of distinct elements in $A$       |
| ★ Power Set Cardinality                | $ \mathcal{P}(A)  = 2^{ A }$                                               | Any finite set of size $n$               |
| ★ Total Subsets                        | $N_{\text{total}} = 2^n$                                                   | Includes $\emptyset$ and $A$             |
| Proper Subsets                         | $N_{\text{proper}} = 2^n - 1$                                              | Excludes $A$ itself                      |
| Non-Empty Subsets                      | $N_{\text{non-empty}} = 2^n - 1$                                           | Excludes $\emptyset$                     |
| Non-Empty Proper Subsets               | $N_{\text{ne-proper}} = 2^n - 2$                                           | Excludes both $\emptyset$ and $A$        |
| Cartesian Product Size                 | $ A \times B  =  A  \cdot  B $                                             | Ordered pairs: $(a, b) \neq (b, a)$      |
| General Cartesian Size                 | $ A_1 \times \dots \times A_k  = \prod_{i=1}^k  A_i $                      | $k$ -ary Cartesian product               |
| Total Binary Relations                 | $ \mathcal{P}(A \times B)  = 2^{ A  \cdot  B }$                            | Number of binary relations               |
| ★ Reflexive Relations Count            | $N_{\text{reflexive}} = 2^{n(n-1)}$                                        | Distinct reflexive relations on size $n$ |
| ★ Total Functions $A \rightarrow B$    | $N_{\text{functions}} =  B ^{ A }$                                         | Unrestricted mappings                    |
| ★ Total Injections (One-to-One)        | $P( B ,  A ) = \frac{ B !}{( B - A )!}$                                    | Requires $ A  \leq  B $                  |
| ★ Total Bijections (Correspondences)   | $N_{\text{bijections}} = n!$                                               | Finite sets with $ A  =  B  = n$         |
| ★ Difference Cardinality               | $ A \setminus B  =  A  -  A \cap B $                                       | Relative complement size                 |
| ★ Symmetric Diff. Size                 | $ A \oplus B  =  A \cup B  -  A \cap B $                                   | Elements in exactly one set              |
| ★ Complement Cardinality               | $ \overline{A}  =  \mathcal{U}  -  A $                                     | Unshaded universe region                 |
| PIE (2 Sets)                           | $ A \cup B  =  A  +  B  -  A \cap B $                                      | Subtracts overlap once                   |
| PIE (3 Sets)                           | $ A \cup B \cup C  = \sum  A_i  - \sum  A_i \cap A_j  +  A \cap B \cap C $ | Full 3-set inclusion-exclusion           |
| Binary Representation Bits             | $b(n) = \lfloor \log_2 n \rfloor + 1$                                      | Bits to store integer $n > 0$            |

### 9.3 Set Identities and Algebraic Laws

| Set Identities and Algebraic Laws |                                                                                       |                                                        |
|-----------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------|
| Law Name                          | Union Form ( $\cup$ )                                                                 | Intersection Form ( $\cap$ )                           |
| ★ Identity Laws                   | $A \cup \emptyset = A$                                                                | $A \cap \mathcal{U} = A$                               |
| ★ Domination Laws                 | $A \cup \mathcal{U} = \mathcal{U}$                                                    | $A \cap \emptyset = \emptyset$                         |
| ★ Idempotent Laws                 | $A \cup A = A$                                                                        | $A \cap A = A$                                         |
| ★ Double Complement               | $\overline{\overline{A}} = A$                                                         | (Involution Law)                                       |
| ★ Commutative Laws                | $A \cup B = B \cup A$                                                                 | $A \cap B = B \cap A$                                  |
| ★ Associative Laws                | $(A \cup B) \cup C = A \cup (B \cup C)$                                               | $(A \cap B) \cap C = A \cap (B \cap C)$                |
| ★ Distributive Laws               | $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$                                      | $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$       |
| ★ De Morgan's Laws                | $\overline{A \cup B} = \overline{A} \cap \overline{B}$                                | $\overline{A \cap B} = \overline{A} \cup \overline{B}$ |
| ★ Absorption Laws                 | $A \cup (A \cap B) = A$                                                               | $A \cap (A \cup B) = A$                                |
| ★ Complement Laws                 | $A \cup \overline{A} = \mathcal{U}$                                                   | $A \cap \overline{A} = \emptyset$                      |
| ★ Difference Law                  |                                                                                       | $A \setminus B = A \cap \overline{B}$                  |
| ★ Symmetric Difference            | $A \oplus B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$ |                                                        |

— END OF SETS, RELATIONS & FUNCTIONS —